

# Nicholas Eisenberg, Ph.D.

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702-354-0866 || nickeisenberg@gmail.com || [linkedin.com/in/nick-eisenberg](https://www.linkedin.com/in/nick-eisenberg) || [github.com/nickeisenberg](https://github.com/nickeisenberg)

## Experience

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### Senior Data Scientist

*June 2022 - Present*

*Nevada National Security Site*

- Utilized YOLO v5 to develop an object-detection model in PyTorch that achieved a 80% mAP evaluation on a in-house thermal-image vehicle dataset for a R&D project.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Enhanced the scalability of models initially prototyped by our team of Analysts and Data Scientists for training on our internal high-performance computing (HPC) cluster. My role involved refactoring the existing code to enable efficient execution in a distributed and parallel setting.
- Mentored a summer intern in the development and execution of models on a HPC cluster, involving multi-node, multi-GPU setups and Slurm resource management.
- Developed a Django web interface that enabled non-technical staff to interact with the company's Neo4j graph database, allowing them to query and update data without requiring knowledge of the query syntax.

## Technical Skills

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**Programming Languages:** Python, SQL, Bash, Cypher, Lua, L<sup>A</sup>T<sub>E</sub>X

**Libraries:** PyTorch, Scikit-Learn, Pandas, Scipy, Sqlalchemy, Matplotlib, Plotly, Streamlit, Django

**Developer Tools:** Git, AWS, Unix/Linux, Vim, Slurm, Tmux, Neo4j

## Projects

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**iron.nvim Maintainer** (GitHub link)

iron.nvim is an interactive REPL plugin for neovim. I have recently joined the developers of iron.nvim to help maintain the code base, add new features and respond to ongoing issues.

## Education

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**Auburn University - Ph.D.**

*June 2022*

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

**University of Nevada, Las Vegas - B.S./M.S.**

*December 2015 / January 2018*

Major: Applied Mathematics

## Publications

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- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Eisenberg, N., Chen, L. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). <https://doi.org/10.1007/s10959-023-01302-4>